

End-to-End Machine Learning Project Report

Task 3: Pipeline, Optimization, and Deployment

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Dataset: California Housing Dataset (20,640 rows, 8 features)

1. Executive Summary

This report documents the design, development, optimization, and evaluation of an end-to-end machine learning regression pipeline. Utilizing the California Housing dataset containing over 20,000 records, the project encompasses data cleaning, feature engineering, feature scaling, training four distinct regression algorithms, hyperparameter tuning via GridSearchCV, cross-validation, and model serialization for real-world deployment.

2. Dataset & Data Preparation

- **Dataset Source:** California Housing Dataset (scikit-learn repository).
- **Dimensions:** 20,640 rows and 8 numeric attributes (including median income, house age, average rooms, population).
- **Data Cleaning:** Missing values verified and duplicate records removed to ensure data integrity.
- **Feature Engineering:** Created a custom feature `RoomsPerHousehold` by dividing average rooms by average occupancy to enhance predictive performance.
- **Data Scaling & Split:** Features normalized using `StandardScaler` (mean=0, variance=1) and split into 80% training and 20% testing partitions.

3. Model Development & Training

Four distinct machine learning regression models were implemented and evaluated on the test set:

- **Linear Regression:** Baseline parametric model establishing linear dependencies.
- **Decision Tree Regressor:** Non-linear hierarchical partitioning model.
- **Random Forest Regressor:** Ensemble bagging method combining multiple decision trees to minimize overfitting and variance.
- **Gradient Boosting Regressor:** Sequential boosting ensemble method optimized for minimizing prediction residuals.

4. Model Optimization & Validation

- **Hyperparameter Tuning:** Implemented `GridSearchCV` to optimize Random Forest hyperparameters (such as `n_estimators` and `max_depth`).

- **Cross-Validation:** Evaluated model stability using 5-fold cross-validation to prevent data leakage and ensure robust generalization across unseen subsets.
- **Feature Importance Analysis:** Identified primary predictors affecting median house value, highlighting the dominance of median income and geographical parameters.

5. Model Serialization & Deployment

The optimized best-performing model was serialized into a lightweight production-ready file named `best_model.pkl` utilizing the `joblib` library, making it ready for integration into web interfaces and API endpoints.

6. Conclusion

The comprehensive end-to-end workflow successfully demonstrated that ensemble techniques like Random Forest achieve superior predictive accuracy for complex regression tasks. All artifacts, code notebooks, serialized models, and visualizations have been packaged and deployed to GitHub.